

Complete List of Greek Letters in Latex

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The Greek Letters are used widely in $\text{\LaTeX}$ mathematical equation typesetting. Due to a long history of cultural transmission and regional variation, some letters have developed distinct script forms. Overleaf provides an excellent guide [1] on Greek Letters that helped me substantially. However, after discussing with Claude AI, I noticed it incomplete, and I didn't find a complete and satisfying one on the web too. So, it's valuable to create one by myself as reference table for understanding and daily practical writing in LaTeX. That is the following table. The basic information is from the Wikipedia article on the Greek alphabet [2].

No.	Letter	Upper-case	Lower-case	Name	Pronunciation (IPA)
1	A α	A (U+0391)	α <code>\alpha</code>	Alpha	$/\text{'}\text{æ}\text{lf}\text{ə}/$
2	B β	B (U+0392)	β <code>\beta</code>	Beta	US: $/\text{'}\text{b}\text{e}\text{r}\text{t}\text{ə}/$ UK: $/\text{'}\text{b}\text{i}\text{:}\text{t}\text{ə}/$
3	Γ γ	Γ <code>\Gamma</code>	γ <code>\gamma</code>	Gamma	$/\text{'}\text{g}\text{æ}\text{m}\text{ə}/$
4	Δ δ	Δ <code>\Delta</code>	δ <code>\delta</code>	Delta	$/\text{'}\text{d}\text{e}\text{l}\text{t}\text{ə}/$
5	E ϵ ε	E (U+0395)	ϵ <code>\epsilon</code> ε <code>\varepsilon</code>	Epsilon	US: $/\text{'}\text{e}\text{p}\text{s}\text{ə}\text{,l}\text{ɑː}\text{n}/$ UK: $/\text{'}\text{e}\text{p}\text{'s}\text{a}\text{i}\text{l}\text{ə}\text{n}/$
6	Z ζ	Z (U+0396)	ζ <code>\zeta</code>	Zeta	US: $/\text{'}\text{z}\text{e}\text{r}\text{t}\text{ə}/$ UK: $/\text{'}\text{z}\text{i}\text{:}\text{r}\text{t}\text{ə}/$
7	H η	H (U+0397)	η <code>\eta</code>	Eta	US: $/\text{'}\text{e}\text{r}\text{t}\text{ə}/$ UK: $/\text{'}\text{i}\text{:}\text{r}\text{t}\text{ə}/$
8	Θ θ ϑ	Θ <code>\Theta</code>	θ <code>\theta</code> ϑ <code>\vartheta</code>	Theta	US: $/\text{'}\text{\text{θ}}\text{e}\text{r}\text{t}\text{ə}/$ UK: $/\text{'}\text{\text{θ}}\text{i}\text{:}\text{r}\text{t}\text{ə}/$
9	I ι	I (U+0399)	ι <code>\iota</code>	Iota	US: $/\text{a}\text{r}\text{'}\text{o}\text{u}\text{t}\text{ə}/$ UK: $/\text{a}\text{r}\text{'}\text{ə}\text{u}\text{t}\text{ə}/$
10	K κ $\varkappa$	K (U+039A)	κ <code>\kappa</code> $\varkappa$ <code>\varkappa</code>	Kappa	$/\text{'}\text{k}\text{æ}\text{p}\text{ə}/$

No.	Letter	Upper-case	Lower-case	Name	Pronunciation (IPA)
11	$\Lambda \lambda$	Λ <code>\Lambda</code>	λ <code>\lambda</code>	Lambda	$/\text{'læmdə}/$
12	$\mathbf{M} \mu$	$\mathbf{M}$ <code>(U+039C)</code>	μ <code>\mu</code>	Mu	$/\text{mju:}/$
13	$\mathbf{N} \nu$	$\mathbf{N}$ <code>(U+039D)</code>	ν <code>\nu</code>	Nu	US: $/\text{nu:}/$ UK: $/\text{nju:}/$
14	$\Xi \xi$	Ξ <code>\Xi</code>	ξ <code>\xi</code>	Xi	$/\text{zai}/, / \text{sai}/, / \text{ksai}/$
15	$\mathbf{O} \circ$	$\mathbf{O}$ <code>(U+039F)</code>	$\circ$ <code>(U+03BF)</code>	Omicron	US: $/\text{'ɒmɪkrɒn}/$ UK: $/\text{'ɒmɪkrɒn}/$
16	$\Pi \pi \varpi$	Π <code>\Pi</code>	π <code>\pi</code> ϖ <code>\varpi</code>	Pi	$/\text{pai}/$
17	$\mathbf{P} \rho \varrho$	$\mathbf{P}$ <code>(U+03A1)</code>	ρ <code>\rho</code> ϱ <code>\varrho</code>	Rho	US: $/\text{rou}/$ UK: $/\text{rəu}/$
18	$\Sigma \sigma \varsigma$	Σ <code>\Sigma</code>	σ <code>\sigma</code> ς <code>\varsigma</code>	Sigma	$/\text{'sɪgmə}/$
19	$\mathbf{T} \tau$	$\mathbf{T}$ <code>(U+03A4)</code>	τ <code>\tau</code>	Tau	US: $/\text{tau}/, / \text{tə}/$ UK: $/\text{taʊ}/, / \text{tɔ:}/$
20	$\Upsilon \upsilon$	Υ <code>\Upsilon</code>	υ <code>\upsilon</code>	Upsilon	US: $/\text{'ju:psə,lɒn}/, / \text{'ʌp-}/$ UK: $/\text{'ʌpsɪ,lɒn}/$
21	$\Phi \phi \varphi$	Φ <code>\Phi</code>	ϕ <code>\phi</code> φ <code>\varphi</code>	Phi	$/\text{fai}/$
22	$\mathbf{X} \chi$	$\mathbf{X}$ <code>(U+03A7)</code>	χ <code>\chi</code>	Chi	$/\text{kai}/$
23	$\Psi \psi$	Ψ <code>\Psi</code>	ψ <code>\psi</code>	Psi	$/\text{sai}/, / \text{psai}/$
24	$\Omega \omega$	Ω <code>\Omega</code>	ω <code>\omega</code>	Omega	US: $/\text{ou'megə}/$ UK: $/\text{'əʊmɪgə}/$

There are 24 Greek letters, and each one has both Upper and lower cases. The upper-cases of 13 of them—“A, B, E, Z, H, I, K, M, N, O, P, T, X”, and the lower case of letter Omicron—“o”, have identical looking to their corresponding Latin letter, but in fact, they are different unicode characters and defined in Greek and Coptic block (U+0370-U+03FF) [3]([4] for detail) . These 14 (= 13 + 1) unicodes are presented in the upper and lower case columns with the format `(U+XXXX)`, for example, in the row 1, the upper-case is “A `(U+0391)`”, and in row 15, the lower-case is “o `(U+03BF)`”.

The other upper and lower cases of Greek letters, from now on we could call them symbols too, have their distinct forms and were extensively used in math and physics expressions. They can be typeset by LaTeX engines effortlessly using LaTeX

commands. These symbol together with their LaTeX commands are presented in the upper and lower case columns with the format `\ccc`, for example, in the row 1, the lower-case is “ α `\alpha`”, in the row 24, the Upper-case is “ Ω `\Omega`” and the lower-case is “ ω `\omega`”.

While the Greek culture spreading to the world, the letter forms emerge different variants. There are 7 Greek letters whose lower cases have two different symbols, so LaTeX define the variant symbol with a “var” prefix. Those 7 letters are: Epsilon (row 5), Theta (row 8), Kappa (row 10), Pi (row 16), Rho (Row 17), Sigma (row 18), and Phi (row 21).

Totally, there are 41 math symbols defined in LaTeX. The calculation is:

$$\begin{aligned} \# \text{ of symbols} &= 24 \text{ (\# of Greek Letters)} \\ &\quad * 2 \text{ (Upper and lower cases)} \\ &\quad + 7 \text{ (\# of lower case variants)} \\ &= 55 \end{aligned}$$

$$\begin{aligned} \# \text{ of symbols defined in LaTeX} &= 55 \text{ (\# of symbols calculated)} \\ &\quad - 14 \text{ (\# of upper\&lower cases identical to Latin)} \\ &= 41 \end{aligned}$$

40 of 41 is defined natively in LaTeX engine that could be found in the original source code [5]. The special one “ $\varkappa$ `\varkappa`” is defined in the package `amsmath` or `amsfonts` that you could find in the source code [6].

References

- [1] Overleaf. *List of Greek letters and math symbols*. Web Page. Accessed: May 10, 2026. URL: https://www.overleaf.com/learn/latex/List_of_Greek_letters_and_math_symbols.
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- [3] Wikipedia contributors. *Greek script in Unicode*. Web Page. Accessed: Jun 4, 2026. URL: https://en.wikipedia.org/wiki/Greek_script_in_Unicode.
- [4] The Unicode Consortium. *Greek and Coptic: Unicode Character Chart U+0370–U+03FF*. The Unicode Standard, Version 17.0. PDF chart. Accessed: Jun. 4, 2026. 2025. URL: <https://www.unicode.org/charts/PDF/U0370.pdf>.
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- [6] American Mathematical Society. *The amssymb package source: amssymb.dtx*. Version 3.01; part of the AMSFonts source files on CTAN. Jan. 14, 2013. URL: <https://mirrors.rit.edu/CTAN/fonts/amsfonts/source/amssymb.dtx> (visited on 06/04/2026).